

Input Tokens

|   | 小 | 狗 | 和 | 小 | 貓 | 玩 |
|---|---|---|---|---|---|---|
| t | 0 | 1 | 2 | 3 | 4 | 5 |
|   | 0 | 1 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 1 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 1 | 0 | 0 | 1 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 1 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 1 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |
|   | 0 | 0 | 0 | 0 | 0 | 0 |

狗 0  
貓 1  
牛 2  
大 3  
中 4  
小 5  
吃 6  
玩 7  
跑 8  
和 9  
你 10  
我 11  
他 12  
的 13  
好 14  
在 15

Word Embedding

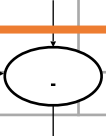
| 狗 | 貓 | 牛 | 大 | 中 | 小 | 吃 | 玩 | 跑 | 和 | 你  | 我  | 他  | 的  | 好  | 在  |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 3 | 0 | 2 | 1 | 2 | 3 | 2 | 2 | 3 | 0 | 1  | 0  | 0  | 0  | 0  | 1  |
| 0 | 3 | 0 | 3 | 0 | 2 | 2 | 1 | 1 | 3 | 0  | 1  | 1  | 0  | 1  | 0  |
| 0 | 1 | 1 | 2 | 3 | 1 | 1 | 3 | 0 | 1 | 1  | 0  | 1  | 1  | 0  | 1  |
| 1 | 1 | 3 | 3 | 3 | 3 | 2 | 1 | 3 | 2 | 1  | 1  | 0  | 1  | 1  | 0  |
| 3 | 1 | 0 | 3 | 0 | 2 | 3 | 2 | 2 | 2 | 0  | 0  | 0  | 1  | 0  | 0  |

→  
→  
→  
→  
→

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ | ↓ | ↓ |
| 3 | 3 | 0 | 3 | 0 | 2 |
| 2 | 0 | 3 | 2 | 3 | 1 |
| 1 | 0 | 1 | 1 | 1 | 3 |
| 3 | 1 | 2 | 3 | 1 | 1 |
| 2 | 3 | 2 | 2 | 1 | 2 |

Positional

t 0 1 2 3 4 5



Encoding

| d |  |        |       |       |       |       |       |
|---|--|--------|-------|-------|-------|-------|-------|
| 0 |  | 0.922  | 0.987 | 0.998 | 1.000 | 1.000 | 1.000 |
| 1 |  | 0.715  | 0.312 | 0.126 | 0.050 | 0.020 | 0.008 |
| 2 |  | 0.368  | 0.889 | 0.982 | 0.997 | 1.000 | 1.000 |
| 3 |  | 1.000  | 0.592 | 0.250 | 0.100 | 0.040 | 0.016 |
| 4 |  | -0.408 | 0.702 | 0.951 | 0.992 | 0.999 | 1.000 |

|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| 3.922 | 3.987 | 0.998 | 4.000 | 1.000 | 3.000 |
| 2.715 | 0.312 | 3.126 | 2.050 | 3.020 | 1.008 |
| 1.368 | 0.889 | 1.982 | 1.997 | 2.000 | 4.000 |
| 4.000 | 1.592 | 2.250 | 3.100 | 1.040 | 1.016 |
| 1.592 | 3.702 | 2.951 | 2.992 | 1.999 | 3.000 |



# Transformer Encoder

## Multi-head Self-Attention

| Head | Wq |    |    |    |    |   |
|------|----|----|----|----|----|---|
| 0    | -1 | -1 | 1  | 1  | 1  | → |
|      | 1  | -1 | -1 | 0  | -1 | → |
|      | -1 | -1 | 0  | 0  | 1  | → |
| 1    | 0  | 1  | 1  | 1  | 1  | → |
|      | 0  | 1  | -1 | 1  | 1  | → |
|      | -1 | 1  | 0  | -1 | 0  | → |

|   |  | Wk |    |    |    |    |   |
|---|--|----|----|----|----|----|---|
| 0 |  | -1 | -1 | 1  | 1  | 1  | → |
|   |  | 1  | -1 | -1 | 0  | -1 | → |
|   |  | -1 | -1 | 0  | 0  | 1  | → |
| 1 |  | 0  | 1  | -1 | -1 | 1  | → |
|   |  | 0  | 1  | -1 | 0  | 0  | → |
|   |  | -1 | 1  | 0  | -1 | 0  | → |

|   |  |    |    |    |   |    |   |
|---|--|----|----|----|---|----|---|
|   |  | Wv |    |    |   |    |   |
| 0 |  | 1  | 1  | -1 | 0 | 1  | → |
|   |  | 0  | 1  | 0  | 1 | 0  | → |
|   |  | 0  | -1 | -1 | 1 | -1 | → |
| 1 |  | -1 | 1  | 1  | 0 | 1  | → |
|   |  | -1 | 0  | 0  | 1 | -1 | → |

|        |        |        |        |        |        |  |  |      |
|--------|--------|--------|--------|--------|--------|--|--|------|
|        |        |        |        |        |        |  |  | Head |
| Q      |        |        |        |        |        |  |  | 0    |
| 0.323  | 1.884  | 3.059  | 2.040  | 1.018  | 4.008  |  |  |      |
| -1.753 | -0.915 | -7.061 | -3.040 | -6.018 | -5.008 |  |  |      |
| -5.044 | -0.597 | -1.173 | -3.058 | -2.021 | -1.008 |  |  |      |
| 9.675  | 6.495  | 10.308 | 10.140 | 8.058  | 9.024  |  |  |      |
| 6.939  | 4.717  | 6.344  | 6.145  | 4.059  | 1.024  |  |  |      |
| -5.207 | -5.268 | -0.122 | -5.050 | 0.980  | -3.008 |  |  |      |

| K      |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.323  | 1.884  | 3.059  | 2.040  | 1.018  | 4.008  |
| -1.753 | -0.915 | -7.061 | -3.040 | -6.018 | -5.008 |
| -5.044 | -0.597 | -1.173 | -3.058 | -2.021 | -1.008 |
| -1.060 | 1.532  | 1.845  | -0.055 | 1.979  | -1.008 |
| 1.347  | -0.577 | 1.144  | 0.053  | 1.020  | -2.992 |
| -5.207 | -5.268 | -0.122 | -5.050 | 0.980  | -3.008 |

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
|        |        |        |        |        |        |
| V      |        |        |        |        |        |
| 6.861  | 7.112  | 5.092  | 7.045  | 4.019  | 3.008  |
| 6.714  | 1.904  | 5.376  | 5.151  | 4.060  | 2.024  |
| -1.675 | -3.311 | -5.809 | -3.939 | -5.978 | -6.992 |
| 1.753  | 0.915  | 7.061  | 3.040  | 6.018  | 5.008  |
| -1.515 | -6.097 | -1.699 | -3.892 | -1.959 | -4.984 |

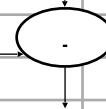
|   |   |    |    |   |   |
|---|---|----|----|---|---|
| 0 | 0 | -1 | -1 | 0 | → |
|---|---|----|----|---|---|

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| -5.367 | -2.481 | -4.232 | -5.097 | -3.040 | -5.016 |
|--------|--------|--------|--------|--------|--------|

|        |
|--------|
| Linear |
|--------|

|    |    |    |    |    |    |   |
|----|----|----|----|----|----|---|
| 0  | 0  | 1  | 1  | 0  | -1 | → |
| 1  | 1  | 1  | 1  | 1  | -1 | → |
| -1 | -1 | -1 | -1 | -1 | 1  | → |
| -1 | 0  | -1 | 1  | 0  | 1  | → |
| -1 | -1 | -1 | -1 | 0  | 1  | → |

## Add & Norm



|         |        |         |        |         |        |
|---------|--------|---------|--------|---------|--------|
| 5.552   | 1.229  | 5.211   | 1.547  | 4.862   | 0.581  |
| 11.729  | -0.386 | 15.952  | 3.561  | 15.406  | 2.854  |
| -7.646  | 1.587  | -10.844 | 0.487  | -10.387 | 2.153  |
| -2.600  | 1.990  | 5.884   | 2.462  | 4.724   | 0.247  |
| -13.508 | -1.638 | -11.722 | -4.610 | -12.275 | -4.933 |

|  |  |           |
|--|--|-----------|
|  |  |           |
|  |  | Normalize |

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.678  | 0.442  | 0.362  | 0.270  | 0.378  | 0.131  |
| 1.290  | -0.619 | 1.264  | 0.905  | 1.285  | 0.875  |
| -0.629 | 0.676  | -0.986 | -0.064 | -0.933 | 0.645  |
| -0.129 | 0.941  | 0.419  | 0.559  | 0.366  | 0.022  |
| -1.210 | -1.440 | -1.060 | -1.670 | -1.096 | -1.673 |

|  |   |   |   |   |   |   |
|--|---|---|---|---|---|---|
|  | 1 | 1 | 1 | 1 | 1 | 1 |
|  | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ |

|               |
|---------------|
| Scale & Shift |
| gamma         |

|  |      |
|--|------|
|  | beta |
|--|------|

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 2 | → |
| 0 | 2 | 0 | 0 | 0 | 3 | → |
| 0 | 0 | 5 | 0 | 0 | 0 | → |
| 0 | 0 | 0 | 3 | 0 | 1 | → |
| 0 | 0 | 0 | 0 | 5 | 2 | → |

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 2.678  | 2.442  | 2.362  | 2.270  | 2.378  | 2.131  |
| 5.580  | 1.763  | 5.528  | 4.810  | 5.570  | 4.749  |
| -3.145 | 3.382  | -4.929 | -0.319 | -4.667 | 3.227  |
| 0.612  | 3.822  | 2.256  | 2.676  | 2.098  | 1.065  |
| -4.048 | -5.202 | -3.298 | -6.351 | -3.478 | -6.364 |
| 1      | 1      | 1      | 1      | 1      | 1      |

Linear

|    |    |    |    |    |    |   |
|----|----|----|----|----|----|---|
| -1 | -1 | 1  | 1  | -1 | 0  | → |
| 1  | 0  | 1  | 0  | 1  | 0  | → |
| -1 | 1  | -1 | 1  | 1  | -1 | → |
| 0  | 0  | 0  | -1 | 0  | -1 | → |
| -1 | 0  | 0  | 1  | -1 | 0  | → |
| 0  | 1  | 0  | 1  | 1  | 1  | → |
| 1  | 0  | 1  | 1  | 1  | 0  | → |
| -1 | 0  | 0  | 1  | 0  | -1 | → |
| -1 | 0  | 1  | -1 | 0  | 1  | → |
| 0  | 0  | 0  | -1 | 1  | -1 | → |
| 1  | 1  | 0  | 0  | -1 | 0  | → |
| 1  | 0  | 0  | 1  | -1 | 0  | → |

Feed-Forward

|        |         |        |         |        |        |
|--------|---------|--------|---------|--------|--------|
| ↓      | ↓       | ↓      | ↓       | ↓      | ↓      |
| -6.743 | 8.201   | -7.265 | 1.627   | -7.038 | 3.775  |
| -4.515 | 0.622   | -5.864 | -4.400  | -5.767 | -1.006 |
| 1.611  | -6.440  | 6.054  | -1.815  | 5.478  | -6.908 |
| -1.612 | -4.822  | -3.256 | -3.676  | -3.098 | -2.065 |
| 1.982  | 6.582   | 3.192  | 6.756   | 3.199  | 5.298  |
| 3.144  | 1.384   | 5.487  | 2.136   | 5.190  | 0.450  |
| -3.903 | 4.444   | -3.608 | -1.723  | -3.669 | 0.059  |
| -3.066 | 0.381   | -1.106 | -0.594  | -1.280 | -2.066 |
| -5.436 | -1.882  | -8.548 | -4.266  | -8.143 | 1.031  |
| -5.660 | -10.024 | -6.554 | -10.027 | -6.577 | -8.429 |
| 12.306 | 9.406   | 11.189 | 13.431  | 11.426 | 13.244 |
| 7.338  | 11.466  | 7.917  | 11.297  | 7.955  | 9.560  |

ReLU

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.000  | 8.201  | 0.000  | 1.627  | 0.000  | 3.775  |
| 0.000  | 0.622  | 0.000  | 0.000  | 0.000  | 0.000  |
| 1.611  | 0.000  | 6.054  | 0.000  | 5.478  | 0.000  |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| 1.982  | 6.582  | 3.192  | 6.756  | 3.199  | 5.298  |
| 3.144  | 1.384  | 5.487  | 2.136  | 5.190  | 0.450  |
| 0.000  | 4.444  | 0.000  | 0.000  | 0.000  | 0.059  |
| 0.000  | 0.381  | 0.000  | 0.000  | 0.000  | 0.000  |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 1.031  |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| 12.306 | 9.406  | 11.189 | 13.431 | 11.426 | 13.244 |
| 7.338  | 11.466 | 7.917  | 11.297 | 7.955  | 9.560  |
| 1      | 1      | 1      | 1      | 1      | 1      |

Linear

|    |    |   |   |    |    |    |    |   |    |   |    |    |   |
|----|----|---|---|----|----|----|----|---|----|---|----|----|---|
| 0  | 1  | 0 | 0 | 0  | 0  | 1  | -1 | 1 | -1 | 0 | -1 | 0  | → |
| 0  | -1 | 0 | 0 | 0  | 0  | 1  | -1 | 1 | 0  | 0 | -1 | -1 | → |
| -1 | 1  | 0 | 1 | 1  | -1 | 0  | 1  | 1 | -1 | 0 | 1  | 0  | → |
| 1  | 1  | 1 | 1 | -1 | 1  | -1 | -1 | 1 | 0  | 0 | -1 | -1 | → |
| 0  | -1 | 0 | 1 | 0  | -1 | 1  | 1  | 0 | 1  | 0 | 1  | -1 | → |

ReLU

|       |       |       |        |       |        |
|-------|-------|-------|--------|-------|--------|
| 0.000 | 0.000 | 0.000 | 0.000  | 0.000 | 0.000  |
| 0.000 | 0.000 | 0.000 | 0.000  | 0.000 | 0.000  |
| 6.176 | 9.466 | 5.621 | 14.291 | 5.964 | 11.663 |

|  |       |        |       |       |       |       |
|--|-------|--------|-------|-------|-------|-------|
|  | 0.000 | 0.000  | 0.000 | 0.000 | 0.000 | 0.000 |
|  | 3.194 | 13.285 | 1.429 | 8.162 | 1.765 | 8.168 |

Add & Norm

Normalize

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 0.191  | -0.721 | 0.210  | -0.558 | 0.173  | -0.487 |
| 1.370  | -0.867 | 1.384  | -0.059 | 1.398  | -0.031 |
| 0.334  | 1.518  | -0.408 | 1.742  | -0.241 | 1.735  |
| -0.649 | -0.424 | 0.171  | -0.478 | 0.066  | -0.673 |
| -1.245 | 0.493  | -1.357 | -0.648 | -1.396 | -0.544 |
| 1      | 1      | 1      | 1      | 1      | 1      |
| ↓      | ↓      | ↓      | ↓      | ↓      | ↓      |
| 2.381  | 0.559  | 2.421  | 0.885  | 2.347  | 1.026  |
| 4.370  | 2.133  | 4.384  | 2.941  | 4.398  | 2.969  |
| -0.668 | -3.036 | -0.817 | -3.485 | -0.483 | -3.471 |
| 0.351  | 0.576  | 1.171  | 0.522  | 1.066  | 0.327  |
| -0.490 | 2.986  | -0.714 | 0.704  | -0.793 | 0.912  |

| Scale & Shift |   |   |   |   |      |   |
|---------------|---|---|---|---|------|---|
| gamma         |   |   |   |   | beta |   |
| 2             | 0 | 0 | 0 | 0 | 2    | → |
| 0             | 1 | 0 | 0 | 0 | 3    | → |
| 0             | 0 | 2 | 0 | 0 | 0    | → |
| 0             | 0 | 0 | 1 | 0 | 1    | → |
| 0             | 0 | 0 | 0 | 2 | 2    | → |

## Word Embedding

| [S] | a | and | apples | orings | by | cat | day | dog |
|-----|---|-----|--------|--------|----|-----|-----|-----|
| 0   | 1 | 2   | 3      | 4      | 5  | 6   | 7   | 8   |
| 3   | 0 | 2   | 1      | 2      | 3  | 2   | 2   | 3   |
| 0   | 3 | 0   | 3      | 0      | 2  | 2   | 1   | 1   |
| 0   | 1 | 1   | 2      | 3      | 1  | 1   | 3   | 0   |
| 1   | 1 | 3   | 3      | 3      | 3  | 2   | 1   | 3   |
| 3   | 1 | 0   | 3      | 0      | 2  | 3   | 2   | 2   |
| 1   | 3 | 2   | 1      | 1      | 2  | 1   | 2   | 1   |



|                    |   |   |   |   |   |
|--------------------|---|---|---|---|---|
| K^T * Q / sqrt(dk) |   |   |   |   |   |
| ↓                  | ↓ | ↓ | ↓ | ↓ | ↓ |

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| t | 0 | 1 | 2 | 3 | 4 | 5 |
|   |   |   |   |   |   |   |

|        |        |        |   |        |        |         |        |         |        |
|--------|--------|--------|---|--------|--------|---------|--------|---------|--------|
| -1.060 | 1.347  | -5.207 | → | 15.128 | 15.530 | -1.010  | 13.753 | -4.722  | 4.315  |
| 1.532  | -0.577 | -5.268 | → | 22.083 | 20.197 | 7.376   | 22.281 | 2.795   | 16.791 |
| 1.845  | 1.144  | -0.122 | → | 15.252 | 10.403 | 15.176  | 15.212 | 11.194  | 10.498 |
| -0.055 | 0.053  | -5.050 | → | 15.085 | 15.297 | 0.222   | 14.588 | -2.989  | 8.514  |
| 1.979  | 1.020  | 0.980  | → | 12.197 | 7.220  | 15.448  | 12.350 | 12.154  | 9.212  |
| -1.008 | -2.992 | -3.008 | → | -8.575 | -2.780 | -16.747 | -7.748 | -13.404 | -1.797 |

|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| 0.001 | 0.009 | 0.000 | 0.000 | 0.000 | 0.000 |
| 0.997 | 0.983 | 0.000 | 0.998 | 0.000 | 0.997 |
| 0.001 | 0.000 | 0.432 | 0.001 | 0.277 | 0.002 |
| 0.001 | 0.007 | 0.000 | 0.000 | 0.000 | 0.000 |
| 0.000 | 0.000 | 0.567 | 0.000 | 0.723 | 0.000 |
| 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |

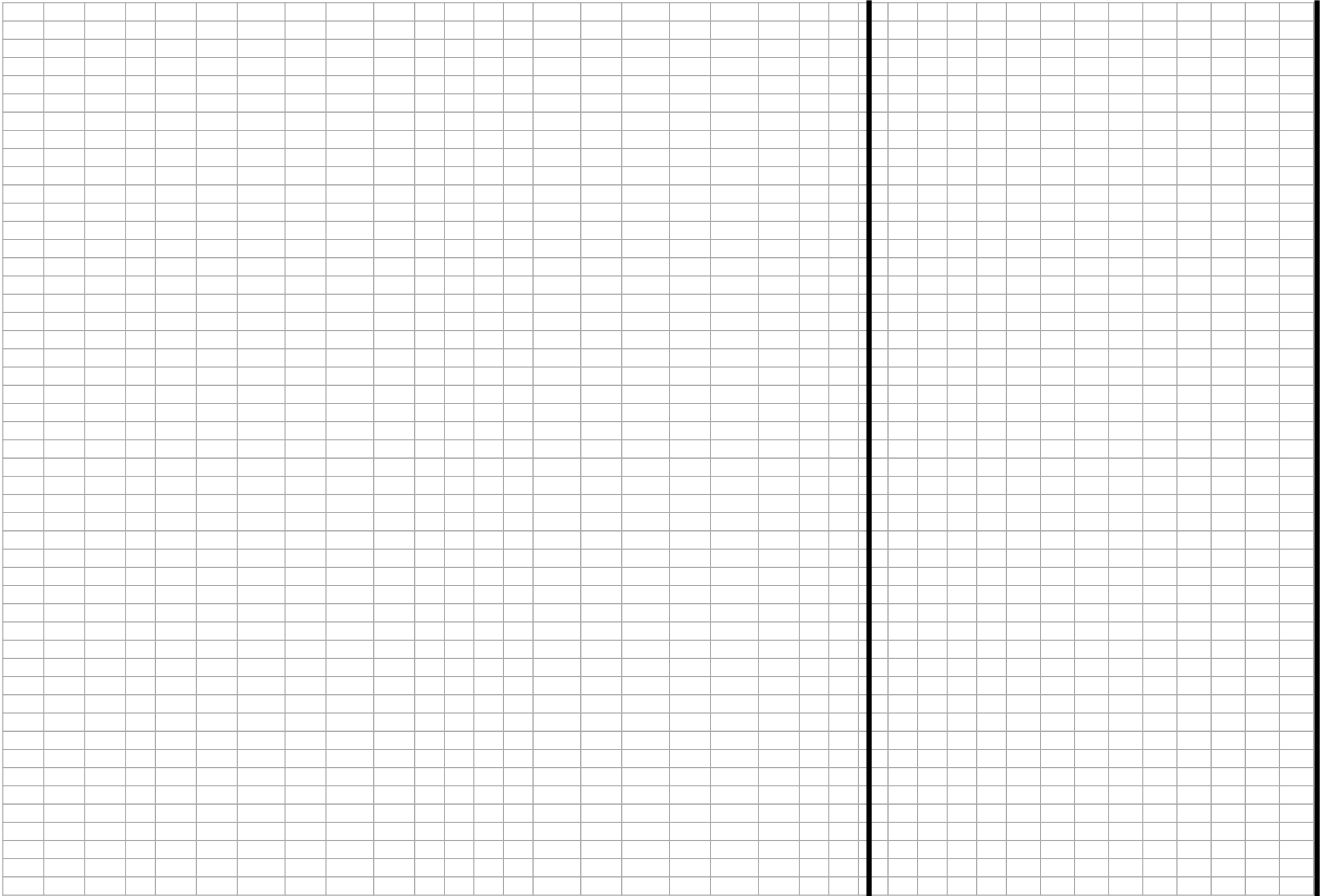
|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| V      |        |        |        |        |        |
| 1.753  | 0.915  | 7.061  | 3.040  | 6.018  | 5.008  |
| -1.515 | -6.097 | -1.699 | -3.892 | -1.959 | -4.984 |
| -5.367 | -2.481 | -4.232 | -5.097 | -3.040 | -5.016 |

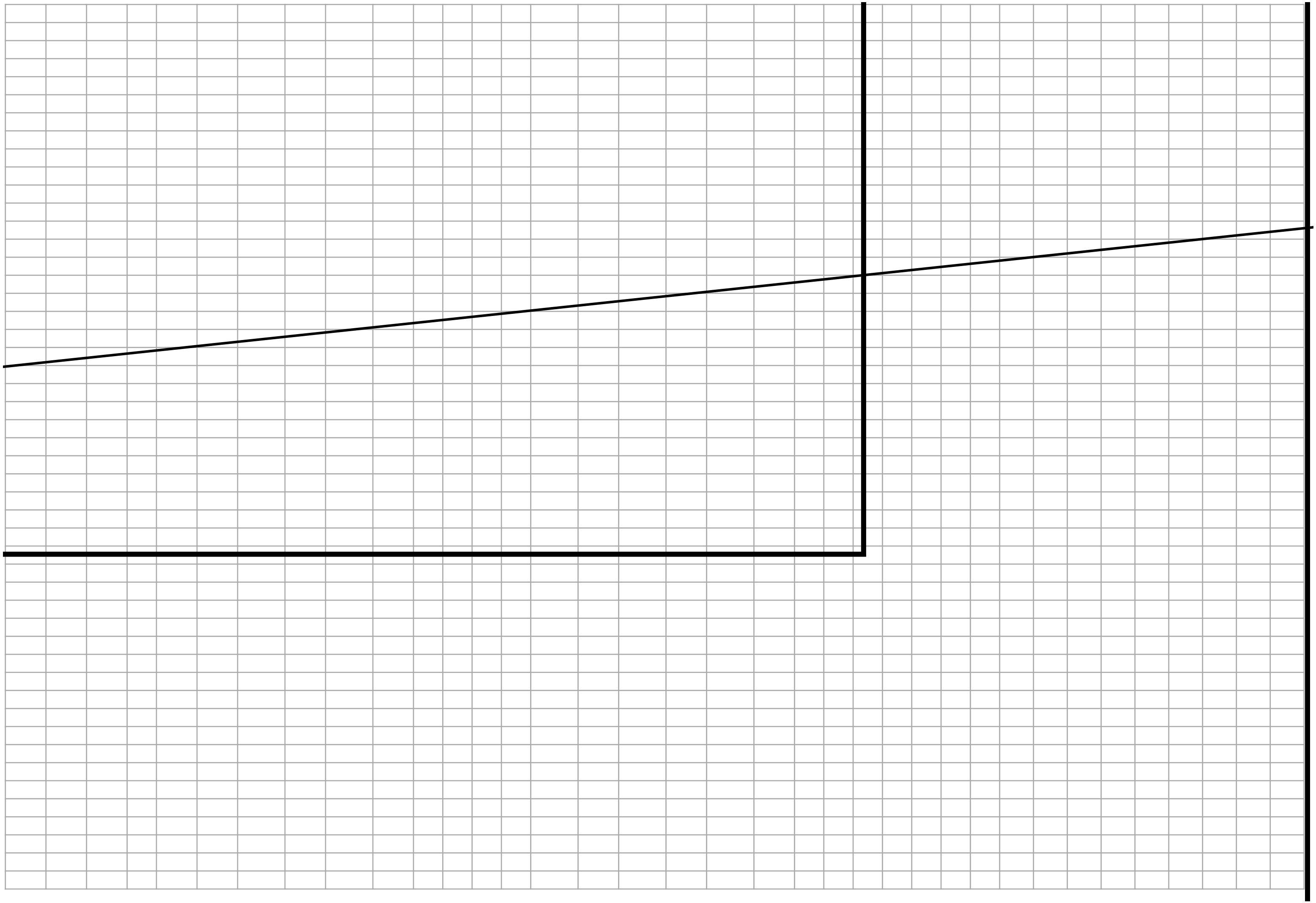
|   |        |        |        |        |        |        |
|---|--------|--------|--------|--------|--------|--------|
|   | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      |
| → | 0.925  | 0.939  | 6.468  | 0.922  | 6.306  | 0.930  |
| → | -6.086 | -6.038 | -1.847 | -6.091 | -1.887 | -6.083 |
| → | -2.488 | -2.527 | -3.555 | -2.485 | -3.369 | -2.483 |

|        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|
| 6.819  | 4.239  | 5.089  | 4.935  | 5.067  | 5.048  |
| 6.650  | 3.860  | 5.371  | 5.120  | 5.344  | 5.304  |
| -1.783 | -6.224 | -5.810 | -5.859 | -5.814 | -5.834 |
| 0.925  | 0.939  | 6.468  | 0.922  | 6.306  | 0.930  |
| -6.086 | -6.038 | -1.847 | -6.091 | -1.887 | -6.083 |
| -2.488 | -2.527 | -3.555 | -2.485 | -3.369 | -2.483 |

|         |        |         |        |         |        |
|---------|--------|---------|--------|---------|--------|
| ↓       | ↓      | ↓       | ↓      | ↓       | ↓      |
| 1.630   | -2.758 | 4.213   | -2.453 | 3.862   | -2.419 |
| 9.014   | -0.698 | 12.826  | 1.510  | 12.386  | 1.847  |
| -9.014  | 0.698  | -12.826 | -1.510 | -12.386 | -1.847 |
| -6.600  | 0.397  | 3.634   | -0.638 | 3.684   | -0.769 |
| -15.100 | -5.341 | -14.673 | -7.602 | -14.273 | -7.938 |







|                      |
|----------------------|
| <b>Output Tokens</b> |
| (shifted right)      |

|          |  |    |  |  |   | [S] | the | dog | and | cat | play |   |   |
|----------|--|----|--|--|---|-----|-----|-----|-----|-----|------|---|---|
|          |  |    |  |  | t | 0   | 1   | 2   | 3   | 4   | 5    | 6 | 7 |
| [S]      |  | 0  |  |  |   | 1   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| a        |  | 1  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| and      |  | 2  |  |  |   | 0   | 0   | 0   | 1   | 0   | 0    | 0 | 0 |
| apples   |  | 3  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| brings   |  | 4  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| by       |  | 5  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| cat      |  | 6  |  |  |   | 0   | 0   | 0   | 0   | 1   | 0    | 0 | 0 |
| day      |  | 7  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| dog      |  | 8  |  |  |   | 0   | 0   | 1   | 0   | 0   | 0    | 0 | 0 |
| eat      |  | 9  |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| fill     |  | 10 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| friend   |  | 11 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| happy    |  | 12 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| house    |  | 13 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| laughter |  | 14 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| light    |  | 15 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| music    |  | 16 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| night    |  | 17 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| play     |  | 18 |  |  |   | 0   | 0   | 0   | 0   | 0   | 1    | 0 | 0 |
| quiet    |  | 19 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| smile    |  | 20 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| the      |  | 21 |  |  |   | 0   | 1   | 0   | 0   | 0   | 0    | 0 | 0 |
| to       |  | 22 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |
| water    |  | 23 |  |  |   | 0   | 0   | 0   | 0   | 0   | 0    | 0 | 0 |

[illegible]





Linear

|    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|
| 0  | 0  | 1  | -1 | 0  | -1 | 1  | 0  | -1 |
| 1  | 1  | 1  | 0  | -1 | -1 | 1  | 1  | -1 |
| -1 | -1 | -1 | 0  | -1 | 1  | -1 | -1 | 1  |
| -1 | 0  | -1 | -1 | 0  | 0  | 1  | 0  | 1  |
| -1 | -1 | -1 | 0  | 1  | 1  | -1 | 0  | 1  |
| 1  | 1  | 1  | 0  | 0  | 0  | 1  | -1 | 1  |

→  
→  
→  
→  
→  
→

Add & Norm



|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 13.117  | -16.929 | 5.411   | 4.244   | -8.004  | -11.797 | 1.211   | 1.809   |
| 24.049  | 8.717   | 22.352  | 20.727  | 14.187  | 11.046  | 17.866  | 16.596  |
| -18.404 | -6.171  | -16.195 | -14.213 | -13.424 | -11.027 | -14.090 | -12.641 |
| 4.195   | 10.479  | 11.768  | 17.106  | 9.979   | 9.915   | 3.157   | 1.676   |
| -16.618 | 1.480   | -11.278 | -12.376 | -0.926  | -0.614  | -12.547 | -11.853 |
| 17.671  | 20.307  | 24.079  | 24.101  | 21.484  | 22.994  | 16.871  | 15.752  |

Normalize

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.510  | -1.508 | -0.036 | -0.140 | -0.878 | -1.110 | -0.063 | -0.006 |
| 1.121  | 0.434  | 0.969  | 0.840  | 0.761  | 0.556  | 1.148  | 1.156  |
| -1.253 | -0.693 | -1.319 | -1.237 | -1.278 | -1.054 | -1.176 | -1.143 |
| 0.011  | 0.568  | 0.341  | 0.625  | 0.450  | 0.474  | 0.078  | -0.017 |
| -1.153 | -0.114 | -1.027 | -1.128 | -0.355 | -0.294 | -1.064 | -1.081 |
| -0.764 | -1.312 | -1.072 | -1.040 | -1.300 | -1.428 | -1.076 | -1.090 |

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 1      | 1      | 1      | 1      | 1      | 1      | 1      | 1      |
| ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      |
| 2.510  | 0.492  | 1.964  | 1.860  | 1.122  | 0.890  | 1.937  | 1.994  |
| 5.242  | 3.869  | 4.938  | 4.680  | 4.522  | 4.112  | 5.296  | 5.313  |
| -5.011 | -2.772 | -5.275 | -4.948 | -5.111 | -4.214 | -4.703 | -4.571 |
| 1.054  | 3.839  | 2.705  | 4.123  | 3.250  | 3.369  | 1.392  | 0.916  |
| -0.612 | -3.546 | -0.108 | -0.511 | -2.580 | -2.823 | -0.254 | -0.323 |
| 5.821  | 8.561  | 7.359  | 7.202  | 8.498  | 9.138  | 7.379  | 7.450  |

Multi-head Cross-Attention

Scale & Shift

gamma

beta

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 2 |
| 0 | 2 | 0 | 0 | 0 | 0 | 3 |
| 0 | 0 | 4 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 5 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 4 | 0 | 4 |
| 0 | 0 | 0 | 0 | 0 | 5 | 2 |

→  
→  
→  
→  
→  
→

Head

Wq

0

1

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 |

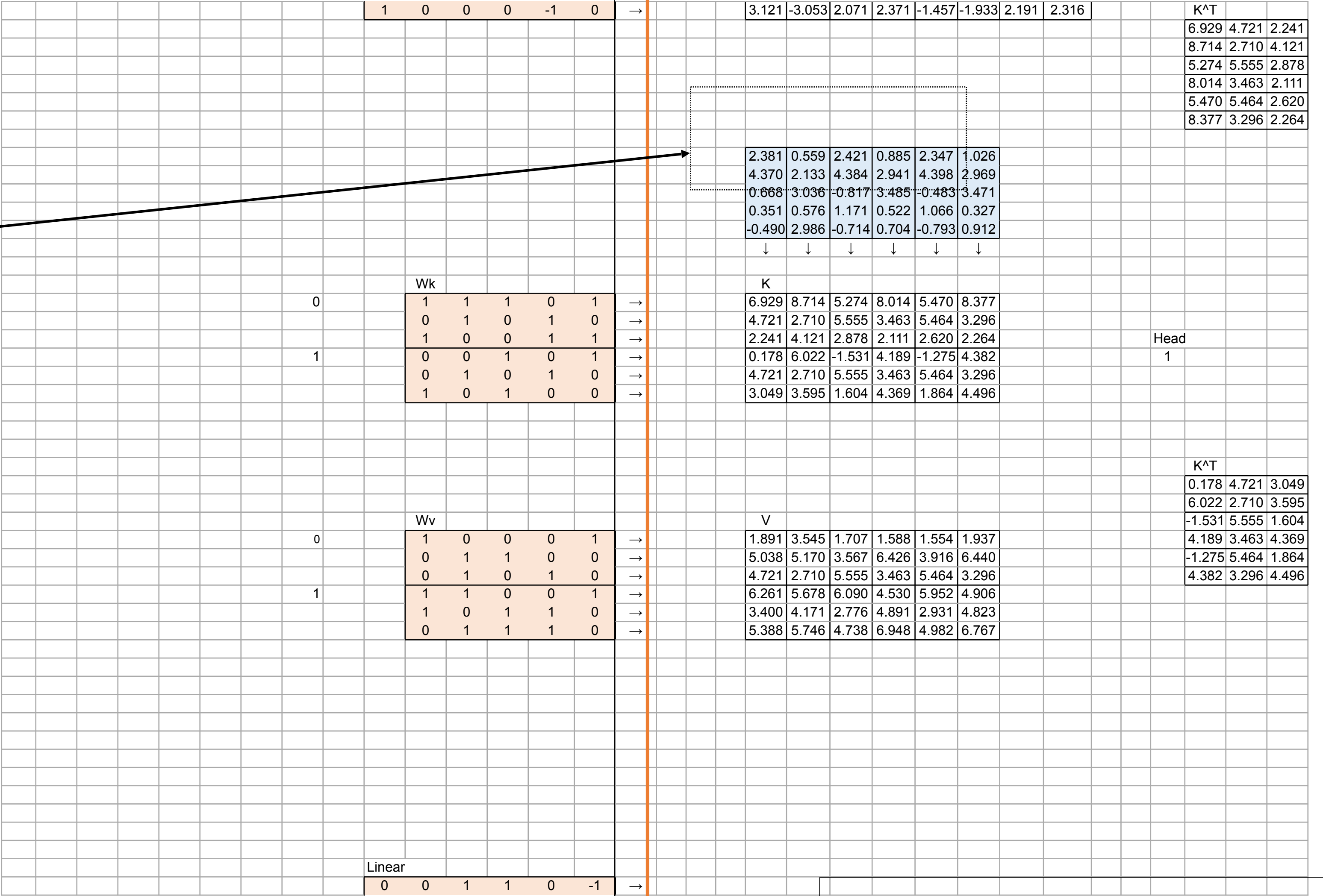
→  
→  
→  
→  
→

Q

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 1.864  | 9.628  | 4.789  | 6.377  | 6.637  | 8.292  | 4.068  | 3.796  |
| 2.510  | 0.492  | 1.964  | 1.860  | 1.122  | 0.890  | 1.937  | 1.994  |
| 5.821  | 8.561  | 7.359  | 7.202  | 8.498  | 9.138  | 7.379  | 7.450  |
| 6.495  | 17.042 | 9.620  | 10.545 | 13.738 | 15.227 | 9.110  | 8.786  |
| 12.118 | 16.269 | 15.002 | 16.004 | 16.270 | 16.618 | 14.067 | 13.679 |

Head  
0

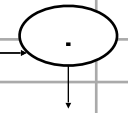




|    |   |    |    |   |   |
|----|---|----|----|---|---|
| 0  | 1 | 1  | 0  | 1 | 0 |
| 0  | 1 | 0  | 0  | 1 | 1 |
| -1 | 0 | 0  | 1  | 0 | 1 |
| 1  | 0 | 0  | -1 | 0 | 1 |
| 1  | 0 | -1 | 0  | 1 | 0 |

→  
→  
→  
→  
→

### Add & Norm



|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 4.005  | 3.134  | 4.384  | 4.348  | 3.763  | 3.532  | 4.315  | 4.312  |
| 17.763 | 15.919 | 17.062 | 16.781 | 16.572 | 16.163 | 17.434 | 17.470 |
| 11.109 | 12.314 | 10.002 | 10.272 | 9.976  | 10.873 | 10.612 | 10.796 |
| 9.094  | 11.718 | 10.600 | 12.013 | 11.130 | 11.248 | 9.290  | 8.818  |
| 4.174  | 7.158  | 3.725  | 3.255  | 6.193  | 6.436  | 3.622  | 3.612  |
| 11.057 | 13.567 | 12.437 | 12.258 | 13.504 | 14.144 | 12.471 | 12.561 |

### Normalize

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| -1.076 | -1.607 | -1.057 | -1.063 | -1.370 | -1.457 | -1.024 | -1.011 |
| 1.601  | 1.132  | 1.463  | 1.352  | 1.361  | 1.223  | 1.506  | 1.507  |
| 0.307  | 0.360  | 0.060  | 0.088  | -0.046 | 0.100  | 0.191  | 0.230  |
| -0.086 | 0.232  | 0.178  | 0.426  | 0.200  | 0.180  | -0.064 | -0.149 |
| -1.043 | -0.745 | -1.188 | -1.276 | -0.852 | -0.841 | -1.157 | -1.145 |
| 0.296  | 0.628  | 0.544  | 0.473  | 0.707  | 0.794  | 0.549  | 0.567  |

1 1 1 1 1 1 1 1

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

|        |       |        |        |        |       |        |        |
|--------|-------|--------|--------|--------|-------|--------|--------|
| 0.924  | 0.393 | 0.943  | 0.937  | 0.630  | 0.543 | 0.976  | 0.989  |
| 6.202  | 5.264 | 5.926  | 5.704  | 5.722  | 5.445 | 6.012  | 6.013  |
| 1.226  | 1.439 | 0.239  | 0.350  | -0.182 | 0.402 | 0.762  | 0.919  |
| 0.572  | 2.160 | 1.892  | 3.129  | 2.002  | 1.900 | 0.678  | 0.257  |
| -0.172 | 1.020 | -0.751 | -1.102 | 0.592  | 0.637 | -0.630 | -0.578 |
| 3.482  | 5.144 | 4.718  | 4.367  | 5.534  | 5.072 | 4.745  | 4.837  |

1 1 1 1 1 1 1 1

### Feed-Forward

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.971  | 1.472  | 1.167  | 2.077  | 2.334  | 1.066  | -0.422 | -1.053 |
| 11.181 | 12.958 | 13.479 | 14.137 | 13.887 | 13.861 | 12.411 | 12.097 |
| 2.040  | 4.047  | 0.370  | -1.451 | 2.312  | 3.568  | 2.223  | 2.932  |
| 5.374  | 3.244  | 5.676  | 5.806  | 4.130  | 3.808  | 5.642  | 5.591  |
| 10.780 | -9.778 | 12.337 | 12.110 | 11.294 | 11.323 | 12.363 | 12.418 |
| -0.666 | 3.336  | -0.720 | -1.089 | 1.221  | 2.566  | -0.135 | 0.165  |
| 11.010 | 13.978 | 12.728 | 13.034 | 14.479 | 14.498 | 11.782 | 11.519 |

### Scale & Shift

| gamma |   |   |   |   |   | beta |  |
|-------|---|---|---|---|---|------|--|
| 1     | 0 | 0 | 0 | 0 | 0 | 2    |  |
| 0     | 2 | 0 | 0 | 0 | 0 | 3    |  |
| 0     | 0 | 4 | 0 | 0 | 0 | 0    |  |
| 0     | 0 | 0 | 5 | 0 | 0 | 1    |  |
| 0     | 0 | 0 | 0 | 4 | 0 | 4    |  |
| 0     | 0 | 0 | 0 | 0 | 5 | 2    |  |

→  
→  
→  
→  
→  
→

### Linear

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| -1 | 1  | -1 | 1  | 1  | -1 | 0  |
| 1  | 1  | 0  | 1  | 0  | 1  | 0  |
| -1 | 0  | 1  | -1 | 1  | 1  | -1 |
| 0  | 1  | 0  | 0  | -1 | 0  | -1 |
| -1 | -1 | 0  | 0  | 1  | -1 | 0  |
| 0  | -1 | 1  | 0  | 1  | 1  | 1  |
| 1  | 1  | 0  | 1  | 1  | 1  | 0  |

→  
→  
→  
→  
→  
→  
→



|    |   |   |   |    |    |    |
|----|---|---|---|----|----|----|
| -1 | 0 | 0 | 0 | 1  | 0  | -1 |
| -1 | 1 | 0 | 1 | -1 | 0  | 1  |
| 0  | 1 | 0 | 0 | -1 | 1  | -1 |
| 1  | 0 | 1 | 0 | 0  | -1 | 0  |
| 1  | 0 | 0 | 0 | 1  | -1 | 0  |

→  
→  
→  
→  
→

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| -2.096 | -0.373 | -2.694 | -3.039 | -1.038 | -0.906 | -2.606 | -2.567 |
| 7.022  | 7.012  | 8.626  | 9.998  | 7.503  | 7.165  | 7.344  | 6.859  |
| 8.856  | 8.385  | 10.394 | 10.173 | 9.664  | 9.780  | 10.387 | 10.429 |
| -1.332 | -3.309 | -3.536 | -3.080 | -5.086 | -5.027 | -3.007 | -2.929 |
| -2.729 | -3.728 | -4.525 | -4.532 | -4.313 | -4.792 | -4.398 | -4.426 |

ReLU

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.971  | 1.472  | 1.167  | 2.077  | 2.334  | 1.066  | 0.000  | 0.000  |
| 11.181 | 12.958 | 13.479 | 14.137 | 13.887 | 13.861 | 12.411 | 12.097 |
| 2.040  | 4.047  | 0.370  | 0.000  | 2.312  | 3.568  | 2.223  | 2.932  |
| 5.374  | 3.244  | 5.676  | 5.806  | 4.130  | 3.808  | 5.642  | 5.591  |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| 0.000  | 3.336  | 0.000  | 0.000  | 1.221  | 2.566  | 0.000  | 0.165  |
| 11.010 | 13.978 | 12.728 | 13.034 | 14.479 | 14.498 | 11.782 | 11.519 |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| 7.022  | 7.012  | 8.626  | 9.998  | 7.503  | 7.165  | 7.344  | 6.859  |
| 8.856  | 8.385  | 10.394 | 10.173 | 9.664  | 9.780  | 10.387 | 10.429 |
| 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| 0      | 0      | 0      | 0      | 0      | 0      | 0      | 0      |
| 1      | 1      | 1      | 1      | 1      | 1      | 1      | 1      |

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓

|         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|
| 20.357  | 25.564  | 24.438  | 26.996  | 26.204  | 25.743  | 21.151  | 20.046  |
| 5.851   | 7.032   | 6.875   | 7.896   | 7.094   | 6.802   | 5.714   | 5.281   |
| 13.751  | 10.022  | 16.220  | 17.691  | 12.301  | 11.422  | 15.011  | 13.954  |
| 14.579  | 17.091  | 15.590  | 17.983  | 15.908  | 16.535  | 14.838  | 15.125  |
| -28.242 | -32.476 | -33.665 | -35.092 | -33.534 | -34.458 | -31.537 | -30.474 |
| 13.059  | 8.314   | 14.320  | 13.877  | 12.164  | 10.660  | 14.399  | 14.277  |

ReLU

|        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|
| 20.357 | 25.564 | 24.438 | 26.996 | 26.204 | 25.743 | 21.151 | 20.046 |
| 5.851  | 7.032  | 6.875  | 7.896  | 7.094  | 6.802  | 5.714  | 5.281  |
| 13.751 | 10.022 | 16.220 | 17.691 | 12.301 | 11.422 | 15.011 | 13.954 |
| 14.579 | 17.091 | 15.590 | 17.983 | 15.908 | 16.535 | 14.838 | 15.125 |
| 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  | 0.000  |
| 13.059 | 8.314  | 14.320 | 13.877 | 12.164 | 10.660 | 14.399 | 14.277 |

Add & Norm

.

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |
|  |  |  |  |  |  |  |  |

Linear

|    |    |   |   |    |    |    |    |    |    |    |    |    |
|----|----|---|---|----|----|----|----|----|----|----|----|----|
| 0  | 1  | 0 | 0 | 0  | 0  | 1  | -1 | 1  | -1 | 0  | -1 | 0  |
| 0  | -1 | 0 | 0 | 0  | 0  | 1  | -1 | 1  | 0  | 0  | -1 | -1 |
| -1 | 1  | 0 | 1 | 1  | -1 | 0  | 1  | 1  | -1 | 0  | 1  | 0  |
| 1  | 1  | 1 | 1 | -1 | 1  | -1 | -1 | 1  | 0  | 0  | -1 | -1 |
| 1  | -1 | 0 | 0 | -1 | 0  | -1 | -1 | -1 | 0  | -1 | 1  | 0  |
| 0  | -1 | 0 | 1 | 0  | -1 | 1  | 1  | 0  | 1  | 0  | 1  | -1 |

→  
→  
→  
→  
→  
→

| Normalize |        |        |        |        |        |        |        |        |
|-----------|--------|--------|--------|--------|--------|--------|--------|--------|
|           | 1.099  | 1.443  | 1.175  | 1.192  | 1.404  | 1.401  | 1.027  | 0.975  |
|           | -0.172 | -0.193 | -0.258 | -0.277 | -0.213 | -0.246 | -0.278 | -0.289 |
|           | 0.230  | -0.293 | 0.158  | 0.178  | -0.293 | -0.296 | 0.230  | 0.176  |
|           | 0.254  | 0.640  | 0.275  | 0.493  | 0.375  | 0.480  | 0.198  | 0.242  |
|           | -1.856 | -1.543 | -1.802 | -1.785 | -1.623 | -1.608 | -1.830 | -1.830 |
|           | 0.446  | -0.054 | 0.452  | 0.199  | 0.350  | 0.268  | 0.653  | 0.726  |
|           | 1      | 1      | 1      | 1      | 1      | 1      | 1      | 1      |
|           | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      |
|           | 3.099  | 3.443  | 3.175  | 3.192  | 3.404  | 3.401  | 3.027  | 2.975  |
|           | 2.655  | 2.614  | 2.483  | 2.445  | 2.574  | 2.508  | 2.443  | 2.422  |
|           | 0.921  | -1.171 | 0.634  | 0.712  | -1.173 | -1.182 | 0.919  | 0.702  |
|           | 2.271  | 4.200  | 2.375  | 3.465  | 2.874  | 3.400  | 1.988  | 2.208  |
|           | -3.425 | -2.172 | -3.208 | -3.140 | -2.491 | -2.432 | -3.318 | -3.319 |
|           | 1      | 1      | 1      | 1      | 1      | 1      | 1      | 1      |
|           | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      | ↓      |

Logit (Z)

## Softmax

[illegible]

| Last Token Position |     |
|---------------------|-----|
| 1                   | 1   |
| 2                   | 2   |
| 3                   | 3   |
| 4                   | 4   |
| 5                   | 5   |
| 6                   | 6   |
| 7                   | 7   |
| 8                   | 8   |
| 9                   | 9   |
| 10                  | 10  |
| 11                  | 11  |
| 12                  | 12  |
| 13                  | 13  |
| 14                  | 14  |
| 15                  | 15  |
| 16                  | 16  |
| 17                  | 17  |
| 18                  | 18  |
| 19                  | 19  |
| 20                  | 20  |
| 21                  | 21  |
| 22                  | 22  |
| 23                  | 23  |
| 24                  | 24  |
| 25                  | 25  |
| 26                  | 26  |
| 27                  | 27  |
| 28                  | 28  |
| 29                  | 29  |
| 30                  | 30  |
| 31                  | 31  |
| 32                  | 32  |
| 33                  | 33  |
| 34                  | 34  |
| 35                  | 35  |
| 36                  | 36  |
| 37                  | 37  |
| 38                  | 38  |
| 39                  | 39  |
| 40                  | 40  |
| 41                  | 41  |
| 42                  | 42  |
| 43                  | 43  |
| 44                  | 44  |
| 45                  | 45  |
| 46                  | 46  |
| 47                  | 47  |
| 48                  | 48  |
| 49                  | 49  |
| 50                  | 50  |
| 51                  | 51  |
| 52                  | 52  |
| 53                  | 53  |
| 54                  | 54  |
| 55                  | 55  |
| 56                  | 56  |
| 57                  | 57  |
| 58                  | 58  |
| 59                  | 59  |
| 60                  | 60  |
| 61                  | 61  |
| 62                  | 62  |
| 63                  | 63  |
| 64                  | 64  |
| 65                  | 65  |
| 66                  | 66  |
| 67                  | 67  |
| 68                  | 68  |
| 69                  | 69  |
| 70                  | 70  |
| 71                  | 71  |
| 72                  | 72  |
| 73                  | 73  |
| 74                  | 74  |
| 75                  | 75  |
| 76                  | 76  |
| 77                  | 77  |
| 78                  | 78  |
| 79                  | 79  |
| 80                  | 80  |
| 81                  | 81  |
| 82                  | 82  |
| 83                  | 83  |
| 84                  | 84  |
| 85                  | 85  |
| 86                  | 86  |
| 87                  | 87  |
| 88                  | 88  |
| 89                  | 89  |
| 90                  | 90  |
| 91                  | 91  |
| 92                  | 92  |
| 93                  | 93  |
| 94                  | 94  |
| 95                  | 95  |
| 96                  | 96  |
| 97                  | 97  |
| 98                  | 98  |
| 99                  | 99  |
| 100                 | 100 |

5

### Next Token Probabilities

|        |  | t  | 0 | 1 | 2 | 3 | 4 | 5     | 6 | 7 |
|--------|--|----|---|---|---|---|---|-------|---|---|
| [S]    |  | 0  |   |   |   |   |   | 0.021 |   |   |
| a      |  | 1  |   |   |   |   |   | 0.000 |   |   |
| and    |  | 2  |   |   |   |   |   | 0.000 |   |   |
| apples |  | 3  |   |   |   |   |   | 0.026 |   |   |
| brings |  | 4  |   |   |   |   |   | 0.007 |   |   |
| by     |  | 5  |   |   |   |   |   | 0.004 |   |   |
| cat    |  | 6  |   |   |   |   |   | 0.000 |   |   |
| day    |  | 7  |   |   |   |   |   | 0.000 |   |   |
| dog    |  | 8  |   |   |   |   |   | 0.000 |   |   |
| eat    |  | 9  |   |   |   |   |   | 0.002 |   |   |
| fill   |  | 10 |   |   |   |   |   | 0.622 |   |   |
| friend |  | 11 |   |   |   |   |   | 0.000 |   |   |



[illegible]















